

Assignment 1: Big Data Analytics

Q1. (a) Differentiate between Hierarchical and K-Means clustering based on computational complexity, shape of clusters, and the requirement of the K parameter.

Q1. (b) When dealing with "Big Data" (e.g., 10 million records), which of these two would you choose and why?

Q2. (a) Explain why feature scaling (Standardization/Normalization) is a "make or break" step for K-Means but less critical for a Decision Tree.

Q2. (b) Describe three distinct strategies for handling missing values in a dataset. Provide a specific example for each strategy in the context of a "Customer Churn" dataset.

Q3. Construct a table comparing Supervised and Unsupervised learning across five dimensions: Objective, Data Labeling, Common Algorithms, Feedback Mechanism, and Real-world Applications.

Q4. Given the following 1-dimensional dataset: $X = \{2, 4, 10, 12, 3, 20, 30, 11\}$

1. Initialize two clusters with centroids $C_1 = 2$ and $C_2 = 10$.
2. Perform two full iterations of the K-Means algorithm.
3. Show the cluster assignments and the updated centroids after each iteration.

Q5. A bank wants to predict if a customer will "Churn" (leave the bank). Based on historical data: $P(\{\text{Churn}\}) = 0.20$

- $P(\{\text{High Spending}\} \{\text{Churn}\}) = 0.70$
- $P(\{\text{High Spending}\} \{\text{No Churn}\}) = 0.30$

If a new customer is a "High Spender," what is the probability they will Churn?

Q6. In a transaction database of a pharmacy, the itemset $\{\{\text{Vitamin C}\}, \{\text{Zinc}\}\}$ appears in 150 transactions. The itemset $\{\text{Vitamin C}\}$ appears in 500 transactions.

1. Calculate the Confidence for the rule: $\{\text{Vitamin C}\} \rightarrow \{\text{Zinc}\}$.
2. If the total number of transactions is 1,000, calculate the Support for the itemset $\{\{\text{Vitamin C}\}, \{\text{Zinc}\}\}$.
3. Explain if this rule is "Strong" if the minimum confidence threshold is 40%.

Q7. Discuss how clustering is applied in the following three industries. For each, specify if the goal is discovery, efficiency, or personalization:

1. Telecommunications: (e.g., Customer Churn and segmentation).
2. Healthcare: (e.g., Identifying disease sub-types from MRI data).
3. Retail/E-commerce: (e.g., Market Basket Analysis).

Q8. Explain the Elbow Method and the Silhouette Score. Draw a sample "Elbow Plot" and identify where the optimal K would be?

Q9- A Big Data node is evaluating a split for a categorical feature DeviceType (values: Mobile, Tablet, Desktop) to predict if a user will click an ad.

Current Node Data:

- Total instances (N): 1000
- Class distribution: 600 Click (C), 400 No-Click (NC)

Proposed Split on DeviceType = Mobile:

- Left Child (Mobile): 400 total (100 C, 300 NC)
- Right Child (Tablet/Desktop): 600 total (500 C, 100 NC)

Calculate **the Information Gain** for this split.

Q10 In a MapReduce environment, three Mappers process data for a node split. You need to calculate the weighted Gini Impurity for the entire dataset based on their local results.

Target Split Point: Salary > \$50k

Mapper	Left Child ($\leq 50k$)	Right Child ($> 50k$)
Mapper 1	50 Yes, 50 No	80 Yes, 20 No
Mapper 2	30 Yes, 70 No	90 Yes, 10 No
Mapper 3	20 Yes, 80 No	70 Yes, 30 No

Task: Calculate the **Total Weighted Gini Impurity** for this split across the cluster.

- *Hint:* Sum the counts from all mappers first to get the global distribution before calculating the Gini for the "Left" and "Right" branches.

Q11- You are clustering server logs based on Request Latency (ms). After the first iteration in a distributed environment, the current centroids are $C_1 = 20$ and $C_2 = 100$.

Data points assigned to C_1 in this iteration: {10, 15, 25, 30, 40}

Data points assigned to C_2 in this iteration: {80, 110, 120, 150}

1. Calculate the new centroids for the next iteration.
2. Calculate the Within-Cluster Sum of Squares (WCSS) for C_1 before the update.
3. Big Data Theory: If you have 10 billion points, how does the Mini-Batch K-Means approach differ numerically from this standard (Batch) approach?

Q12- An engineering firm is designing a Hadoop cluster to store 500 Terabytes (TB) of raw sensor data.

- HDFS Block Size: 128 MB.
 - Replication Factor: 3.
 - DataNodes: Each node has 4 disks of 4 TB each.
 - OS/Log Overhead: 10% of total disk space is reserved for the system.
1. How many physical blocks will be created in the entire cluster for the 500 TB of raw data?
 2. Calculate the Total Physical Storage required in the cluster (including replicas).
 3. Determine the minimum number of DataNodes required to safely store this data.
 4. If the NameNode requires 150 bytes of RAM per block (metadata), how much RAM must the NameNode have to track the raw data and its replicas?

Q13- You need to read a 1 GB file from HDFS.

- The file is split into 128 MB blocks.
 - The network bandwidth between the client and DataNodes is 100 MB/s.
 - The disk read speed is 50 MB/s.
1. Ignoring network overhead/latency, how long will it take to read the first block?
 2. If the client reads the blocks sequentially, what is the total time to read the 1 GB file?

Q14- A streaming service (like Netflix) wants to cluster 50 million users based on their viewing habits (20,000 unique movie tags) to improve recommendations. They use a standard K-Means implementation on a distributed cluster. After the first run, the lead engineer notices two things:

- a) The algorithm takes 15 hours to converge.
 - b) The clusters look "unbalanced"—one cluster has 80% of the users, while others are nearly empty.
1. How might the choice of initial centroids (Random vs. K-Means++) impact the convergence time and the risk of getting stuck in local optima for this specific dataset?
 2. With 20,000 movie tags, the "Distance" between users becomes almost uniform. What is this phenomenon called, and how would Principal Component Analysis (PCA) or Feature Selection improve the K-Means performance?
 3. Suggest how Mini-Batch K-Means could be used here to reduce the 15-hour runtime without losing significant accuracy.

Q15- A financial firm stores transaction logs on an HDFS cluster with a Replication Factor of 3 and a Rack Awareness policy. One afternoon, a Top-of-Rack (ToR) switch fails, taking down an entire rack containing 10 DataNodes.

- a) If the cluster has 3 racks and the failed rack held the "Primary" replica of a specific block, explain the process HDFS uses to ensure that block remains accessible to users.
- b) Once the rack fails, the NameNode detects under-replication. Describe the "Replication Storm" that might occur and how an engineer can configure the `dfs.namenode.replication.work-multiplier-per-iteration` to prevent network saturation.
- c) If a Spark job was scheduled to run on the data in the failed rack, how does the Delay Scheduling algorithm handle this to maintain "Data Locality"?

Q16- You are building a real-time sentiment analysis engine for Twitter using Naïve Bayes. During a major sporting event, a new slang word, "*G.O.A.T.*" (Greatest of All Time), appears frequently in "Positive" tweets. However, this word has zero occurrences in your historical "Negative" or "Neutral" training datasets.

- a) If a tweet contains the word "*G.O.A.T.*", what happens to the total probability of that tweet being "Negative" when using the standard Naïve Bayes formula? Why is this a problem?
- b) Show the numerical adjustment using Laplace Smoothing ($\alpha=1$) for a word count if the total vocabulary size is 50,000.
- c) In Big Data, we often use N-grams (like "not good"). How does moving from Unigrams to Bigrams affect the "Sparsity" of the Naïve Bayes model and the memory requirements of the MapReduce job?

Q17. Discuss the following with examples:

- (a) Bayes' Theorem (Naïve Bayes)
- (b) Naïve Bayes Classifier, Also, explain in what situations Naïve Bayes is not suitable for classification
- (c) Decision Tree
- (d) MapReduce
- (e) YARN